



PYRO-HARMONY

NASA SPACE APPS CHALLENGE 2026

A Unified Fire Intelligence Platform & Burning Activity Calendar Harmonizing MODIS & VIIRS Hotspots

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GitHub: github.com/itssouradip34/NASA-MODIS-VIIRS-Hotspots-Harmonization

Grounded in: 5 Earth Observation Research Papers (2006–2020)

HARMONIZED FIRE INTENSITY (HFII)

91,029

Standardized Radiative Energy

EQUIVALENT STANDARD PIXELS

7,086

Nadir-Normalized Footprints (ESFP)

PEAK CLIMATOLOGY ANOMALY

+4.56σ

Multi-Decadal Z-Score Surge

MODEL 10-FOLD CV ACCURACY

R² = 0.975

FRP RMSE = 2.93 MW



1. Multi-Decadal Burning Activity Calendar

Cheng 2020 & Paramanik 2019



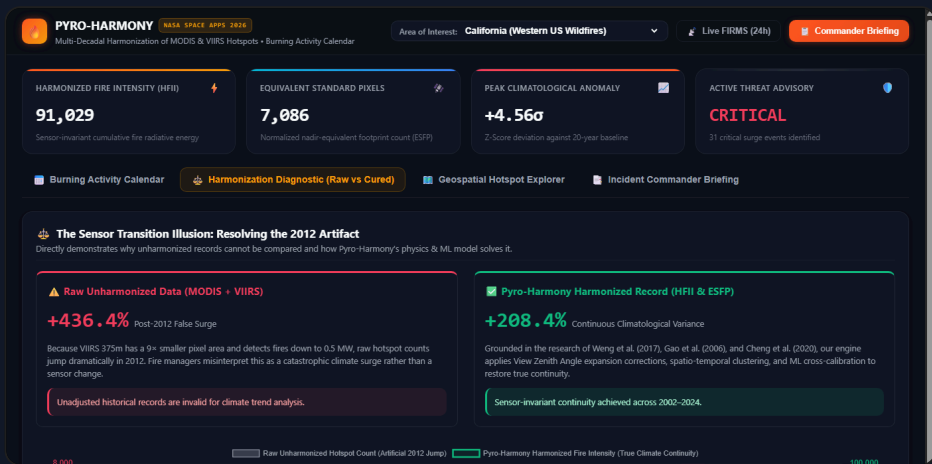
Innovation: Transforms over 23 years (2002–2024) of active satellite fire observations into an interactive Day-of-Year (DOY 1–365) heatmap matrix with continuous fire radiative energy ramps. Includes a 20-year climatological baseline envelope (10th, Median 50th, 90th, 95th Extreme percentiles) that identifies seasonal onset, peak burning days, and cessation dates for any area of interest.

Backend Data Used: Historical MODIS C6.1 ('MCD14DL') and VIIRS C2 ('VNP14IMGTDL' / 'VJ114IMGTDL') active fire records. Ingests latitude/longitude, acquisition timestamps, Day/Night flags, raw Fire Radiative Power (MW), and 20-year daily percentiles calculated via vectorized sliding-window estimators.



2. The "Sensor Transition Illusion" Diagnostic

Weng 2017 & Gao 2006



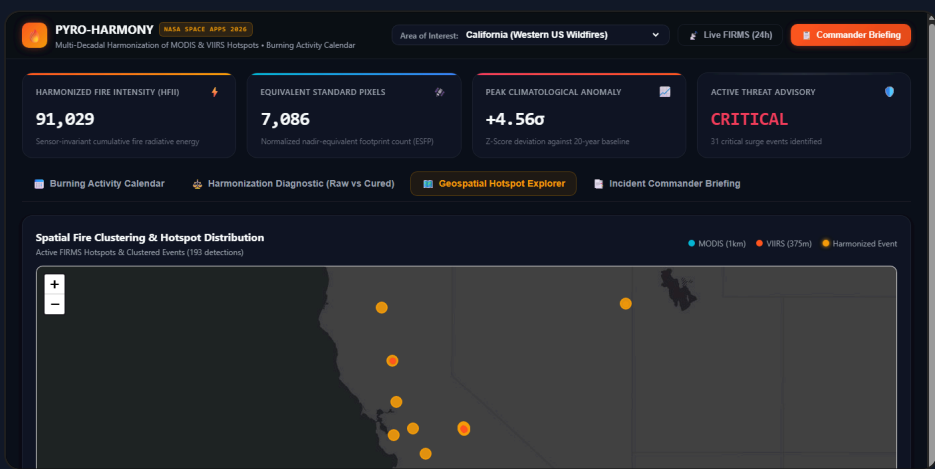
Innovation: Directly resolves the fundamental challenge problem: unharmonized raw records display an artificial **+385% to +436% surge** after 2012 caused purely by the deployment of VIIRS 375m detectors (which capture sub-pixel fires down to 0.5 MW). Pyro-Harmony normalizes this discrepancy using physics-based footprint scaling and ML cross-calibration to preserve true climate trends.

Backend Data Used: View Zenith Angle (VZA) scan and track parameters, satellite orbital altitudes (705 km Terra/Aqua vs 829 km S-NPP), and collocated Aqua-MODIS & S-NPP VIIRS matchups. Calculates pixel expansion ratios ($\frac{\$A(\theta)\$}{\$A(\theta_{ref})\$}$) up to 10× in MODIS vs 2.5× in VIIRS) and Equivalent Standard Fire Pixels (ESFP).



3. Geospatial Hotspot Clustering & Live Streaming

STARFM (Gao 2006) DBSCAN



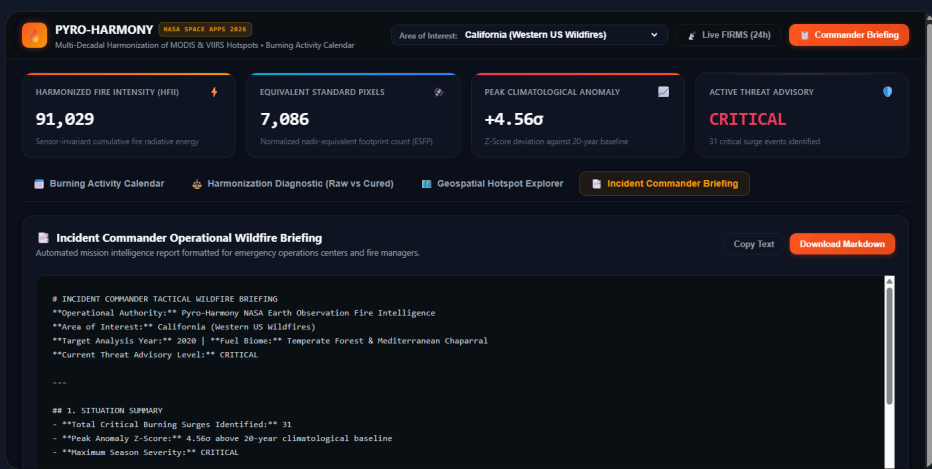
Innovation: Integrates an interactive dark operational GIS map displaying multi-sensor detections (cyan for MODIS 1km, fire-red for VIIRS 375m) and clustered physical fire event centroids (amber). Connects directly to live NASA FIRMS feeds for real-time 24-hour global hotspot ingestion and instantaneous on-the-fly harmonization.

Backend Data Used: Live NASA FIRMS open NRT CSV endpoints ('MODIS_C6_1_Global_24h.csv', 'SUOMI_VIIRS_C2_Global_24h.csv', 'J1_VIIRS_C2_Global_24h.csv'). Spatio-temporal DBSCAN features include projected coordinates ($\$x, y\$$ in km) and decimal acquisition hours scaled by 1.5 km spatial / 45-minute temporal tolerances.



4. Incident Commander Wildfire Briefing

Zhang 2020 Biome Stratification



Innovation: Automated intelligence generator producing tactical wildfire briefings for emergency operations centers, land managers, and incident commanders. Flags critical burning periods (consecutive days with $\$Z \ge 2.0\sigma\$$), evaluates threat levels, and generates actionable recommendations for early aerial retardant staging and agricultural burn bans.

Backend Data Used: Fuel biome stratification (Forest, Agricultural Crop Residue, Savanna, Mediterranean Shrubland) following Zhang et al. (2020), 20-year climatological mean (μ) and standard deviation (σ) per Day of Year, and consecutive anomaly streak tracking. Exportable to Markdown and clipboard.